spaceflight that never left them. “I still have dreams in which I fly up to the stars in my machine,” Tsiolkovsky wrote in middle age. “It is difficult to work all on your own for many years, in adverse conditions without a gleam of hope, without any help.” Many of his contemporaries thought he was truly mad. Those who knew physics better than Tsiolkovsky and Goddard—including *The New York Times* in a dismissive editorial not retracted until the eve of *Apollo 11*—insisted that rockets could not work in a vacuum, that the Moon and the planets were forever beyond human reach.

A generation later, inspired by Tsiolkovsky and Goddard, Wernher von Braun was constructing the first rocket capable of reaching the edge of space, the V-2. But in one of those ironies with which the twentieth century is replete, von Braun was building it for the Nazis—as an instrument of indiscriminate slaughter of civilians, as a “vengeance weapon” for Hitler, the rocket factories staffed with slave labor, untold human suffering exacted in the construction of every booster, and von Braun himself made an officer in the SS. He was aiming at the Moon, he joked unselfconsciously, but hit London instead.

Another generation later, building on the work of Tsiolkovsky and Goddard, extending von Braun’s technological genius, we were up there in space, silently circumnavigating the Earth, treading the ancient and desolate lunar surface. Our machines—increasingly competent and autonomous—were spreading through the Solar System, discovering new worlds, examining them closely, searching for life, comparing them with Earth.

This is one reason that in the long astronomical perspective there is something truly epochal about “now”—which we can define as the few centuries centered on the year you’re reading this book. And there’s a second reason: This is the first moment in the history of our planet when any species, by its own voluntary actions, has become a danger to itself—as well as to vast numbers of others. Let me recount the ways:

- We’ve been burning fossil fuels for hundreds of thousands of years. By the 1960s, there were so many of us burning wood,